CARLO ROMEO

Ph.D. Student

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Portfolio | LinkedIn | GitHub | Google Scholar

PROFESSIONAL SUMMARY

AI Researcher and Machine Learning Engineer currently completing a Ph.D. focused on Reinforcement Learning. My research primarily explores how to make learning algorithms more computationally efficient and capable of operating in data-scarce environments. Drawing from my previous industry experience developing production-level machine learning models, I enjoy bridging the gap between theoretical research and practical application. Motivated by a lifelong love for video games, my goal is to bring this expertise into the gaming industry to develop intelligent agents with complex tactical behaviors and physically simulated locomotion.

WORK EXPERIENCE

Adjunct Professor | *Mediterranean University*, Reggio Calabria, Italy — 2022 – 2023

Lectured on Machine Learning and Deep Learning techniques, utilizing TensorFlow to bridge theoretical concepts with practical application. Evaluated and mentored students on advanced group projects focusing on real-world AI implementations.

Junior Machine Learning Engineer | *Relatech Ithea*, Cosenza, Italy — 04/2021 – 10/2022

Designed and deployed robust AI models as part of an R&D team to meet enterprise customer needs. Engineered and optimized Anomaly Detection and Recommendation Systems for production environments.

Machine Learning Engineer Intern | *Accenture*, Milan, Italy — 10/2020 – 02/2021

Developed and integrated AI solutions directly within the Salesforce ecosystem to drive customer value.

TECHNICAL SKILLS

Frameworks & Tools:	OpenAI Gymnasium, PyTorch, TensorFlow, Keras, JAX, NumPy, Scikit-learn, Pandas, Matplotlib
Programming Languages:	Python, C++, Java
Miscellaneous:	Unity, Unreal Engine, Git, Bash, LaTeX
Languages:	Italian (Native), English (Full Professional Proficiency)

EDUCATION

Ph.D. in Artificial Intelligence | *University of Pisa / University of Florence*, Italy 2022 – 2026

Research focus: Reinforcement Learning, Offline Reinforcement Learning, and Computational Efficiency.

Visiting Ph.D. Researcher: Computer Vision Center (CVC), Universitat Autònoma de Barcelona (05/2025 – 11/2025). Focusing on Token Pruning and Transformers.

M.Sc. in ICT Engineering | *Mediterranean University*, Reggio Calabria, Italy 2018 – 2020

Grade: 110/110 cum laude. Evaluated NLP techniques from a computational perspective and developed a conversational chatbot for resource-constrained systems.

B.Sc. in ICT Engineering | *Mediterranean University*, Reggio Calabria, Italy 2014 – 2018

Grade: 88/110. Developed a First-Person-Shooter video game in Unreal Engine 4, designing complex NPC behavior using state machines.

SELECTED PUBLICATIONS

SPEQ: Offline Stabilization Phases for Efficient Q-Learning in High UTD Ratio RL

Reinforcement Learning Conference (RLC), 2025

This algorithm interleaves low-UTD online training with high-UTD offline stabilization phases to significantly reduce computational overhead. By utilizing dropout regularization for critics, it successfully mitigates overestimation bias while matching the performance of state-of-the-art methods with drastically fewer gradient updates.

NTRL: Encounter Generation via RL for Dynamic Difficulty Adjustment in D&D

Conference on Games (CoG), 2025

NTRL frames combat encounter design as a contextual bandit problem to automate dynamic difficulty adjustment based on real-time party attributes. The approach iteratively optimizes encounters to extend combat longevity and strategic depth, outperforming classic human Dungeon Master heuristics while avoiding total party kills.

Offline Reinforcement Learning with Imputed Rewards

Reinforcement Learning Conference (RLBrew Workshop), 2024

This work proposes a robust Reward Model capable of accurately estimating reward signals from as little as 1% of annotated environment transitions. These imputed rewards enable the successful application of Offline Reinforcement Learning techniques in severely data-scarce scenarios, achieving strong performance on continuous locomotion benchmarks.